Exit Ticket · reflect and retrieve

Exit Ticket

Inserting a linked-list node always requires shifting the stored values of later nodes.

Use key vocabulary from memory

1. Explain node in your own words.

2. Explain head in your own words.

3. Explain Next in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.

Do Now · retrieval

Linked lists using objects

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What makes a heuristic admissible?

Q2

What does a partition do to the pivot?

Q3

What does the head reference identify?

Q4

Why must a new node's Next be set before replacing an existing Next link?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Build a chain — Each Node stores a Value and a Next reference; the final Next is null.

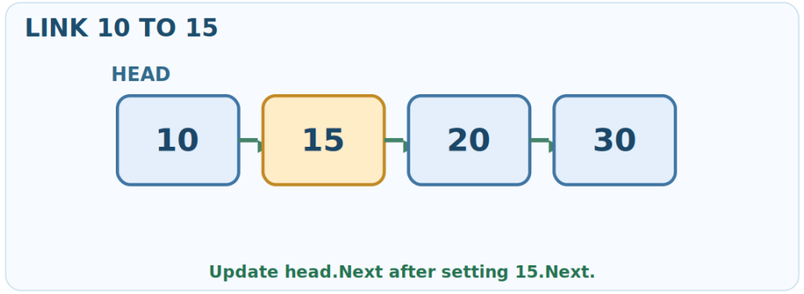
Core idea 2

Preserve the successor — Construct 15 with Next set to the old 20 node *before* changing 10.Next.

Core idea 3

Finish the insertion — Set 10.Next to 15. Now traversal yields 10→15→20→30.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Describe how to delete node 20 from 10→20→30 if you hold a reference to node10. Explain the complexity of reaching the deletion point and changing the link.

Check that you have used the technical terms and explained your decisions.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

For 3→5→8, identify head, 5.Next.Value and 8.Next.

Question 2

Write two C# statements to insert a node with value 12 after node8 in 8→20→30. Give the new traversal.

Check that you have used the technical terms and explained your decisions.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

node — An object containing data and a reference to another node.

head — Reference to the first node.

Next — Link to the successor node.

null — No object is referenced.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

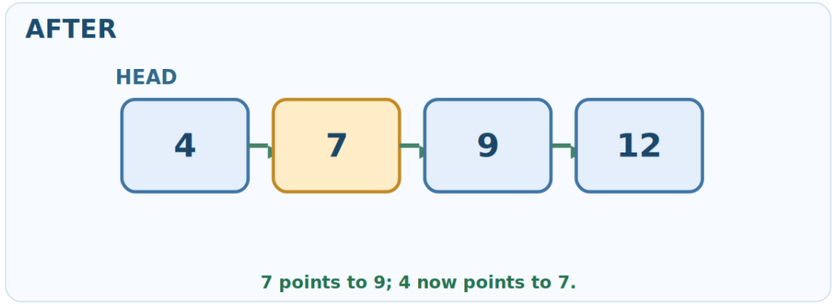
Insert node 7 between 4 and 9 in 4 → 9 → 12.

Follow the worked method

1. Predict first — Set newNode.Next to the existing successor 9.

2. Model the reasoning — Update node4.Next to reference the new 7 node. The 9→12 remainder is preserved.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
First give the new node a reference to 8, the former successor of 2.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.